

UEI604: PROCESS DYNAMICS AND CONTROL

L	T	P	Cr
3	1	2	4.5

Course objective: To make the students understand basic ideas, challenges, techniques, and applications of process control for controlling various processes. To familiarize with different actuators. To classify among different control modes.

Introduction: Historical perspective, Incentives of process control, Synthesis of control system. Classification and definition of process variables.

Mathematical Modeling: Need and application of mathematical modelling, Lumped and distributed parameters, Analogies, Thermal, Electrical and chemical systems, Modeling of Continuous Stirred Tank Reactor (CSTR), Modeling of heat exchanger, Interacting and non-interacting type of systems, Dead time elements

Control Modes: Definition, Characteristics and comparison of on-off, Proportional (P), Integral (I), Differential (D), PI, PD, PID, Dynamic behaviour of feedback controlled processes for different control modes, Control system quality, Integral absolute error (IAE), Integral Squared Error (ISE), Integral Time weighted Absolute Error (ITAE) criterion, Tuning of controllers Ziegler-Nichols, Cohen- Coon methods

Realization of Control Modes: Realization of different control modes like P, I, D, and composite modes in Electronic, Pneumatic controllers.

Actuators: Hydraulic, Pneumatic actuators, Solenoid, I-P and P-I converters, Control valves, Types, Functions, Quick opening, Linear and equal percentage valves, Ball valves, Butterfly valves, Globe valves, Pinch valves, Valve application and selection

Advanced Controls: Introduction to advanced control schemes like Cascade, Feedforward, Ratio, Selective, Override, Split Range and Auctioneering Control.

Laboratory Work: I to P, P to I, Valve characteristics, Simulation of different control modes, Experiments around Basic Process RIG.

Course Learning Outcomes (CLO): After the completion of the course, the student will be able to:

1. Demonstrate a fundamental understanding of process control
2. Develop the mathematical model of various chemical processes
3. Apply the knowledge of different control modes and their applications in controlling various processes
4. Exhibit the knowledge of the working of electronic and pneumatic controllers, and actuators

Text Books:

1. Johnson, C.D., *Process Control Instrumentation Technology*, Prentice Hall of India Private Limited

(2005).

2. Stephanopoulos, G., *Chemical Process Control*, Prentice Hall of India Private Limited (2015).

Reference Books:

1. Harriot, P., *Process Control*, Tata McGraw Hill (1982).
2. Liptak, B.G., *Instrument Engineers Handbook*, Butterworth, Heinemann (2002).
3. Seborg, D.E. and Edgar, T., *Process Dynamics and Control*, John Wiley and Sons (1989).

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST+EST	70
2.	Sessional (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	30